

National University of Computer and Emerging Sciences, Lahore Campus



Course: Analog and Digital Communications
Program: BS(Electrical Engineering)
Duration: 1 Hour
Paper Date: 16-Nov-18
Section: ALL
Exam: MID-II

Course Code: EE323
Semester: Fall 2018
Total Marks: 45
Weight: 20%
Page(s): 2
Roll No:

Instruction/Notes: Attempt all parts of a question together.

Question # 1 [CLO: 4]

[15]

A vestigial shaping filter $H_i(\omega)$ has a transfer function as shown in figure 1. The carrier frequency is $f_c = 100\text{kHz}$ and the baseband signal bandwidth is 5kHz .

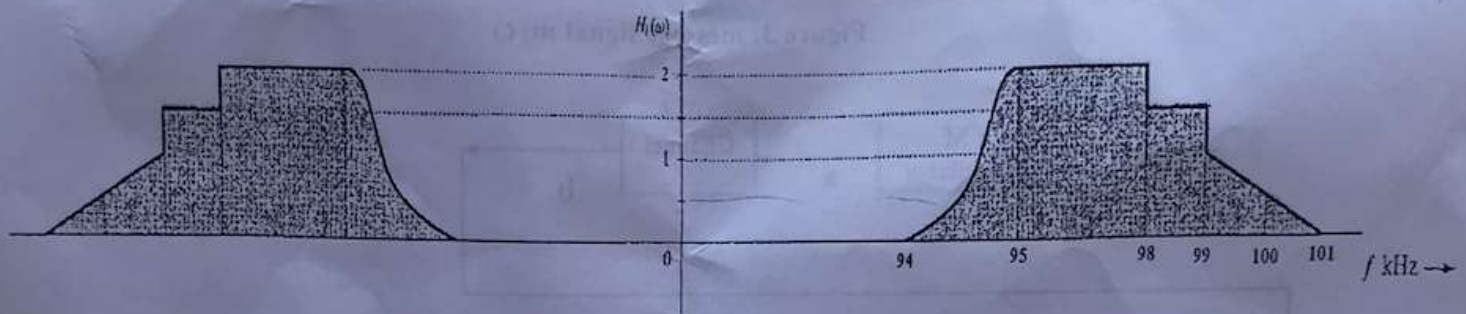


Figure 1: Vestigial shaping filter

- Find the bandwidth of the VSB signal.
- Sketch the necessary equalizer filter response $H_o(\omega)$ for distortion-less reception.
- Design and illustrate a receiver system block diagram for VSB.

Question # 2 [CLO: 5]

[15]

A signal $m(t) = 2 \cos(2\pi * 10^3 t)$ phase modulates (PM) a 1MHz , $2V_{P-P}$ carrier to produce a frequency deviation of 4kHz .

- Write the time-domain expression for the resulting PM signal $\varphi_{PM}(t)$?
- What is the value of phase constant k_p ?
- What is phase deviation $\Delta\phi$?
- What is the bandwidth of the PM signal?
- Can we demodulate $\varphi_{PM}(t)$ using demodulator shown in figure 2, if yes/no explain your answer with reasoning. Suggest modification for given demodulator if required? Also write equation for $e_o(t)$.

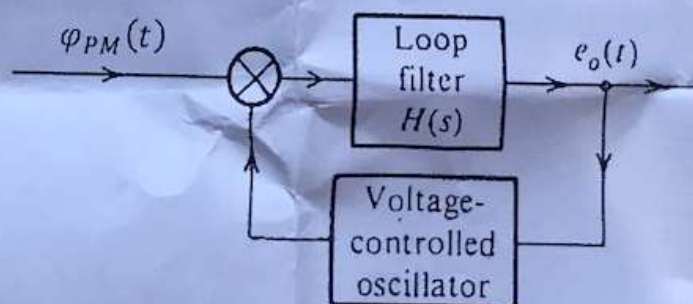


Figure 2: Demodulaor

Question # 3 [CLO: 5]

A message signal $m(t)$, shown in figure 3, frequency-modulates a carrier of frequency $f_c = 10\text{MHz}$ with $k_f = 2\pi * 10^5 \text{Hz}$. The carrier amplitude is A . When the modulated signal is passed through the channel its amplitude is changed randomly. The received signal is demodulated by passing through slope detector as shown in figure 4.

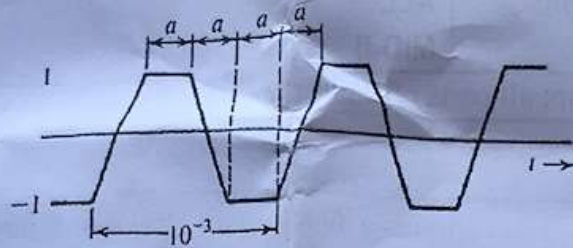
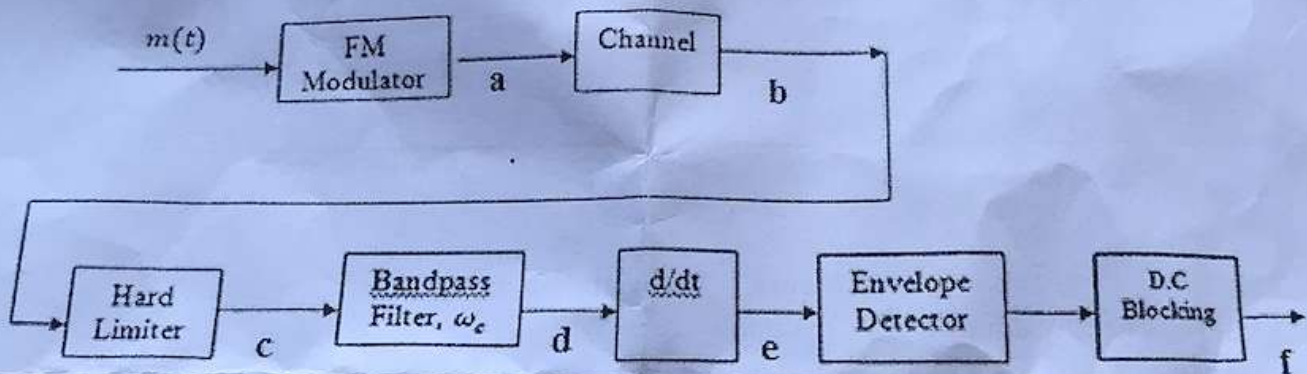
Figure 3: message signal $m(t)$ 

Figure 4: FM modulator and demodulator

- Sketch the waveform, approximated, at points a, b, c, d, e and f.
- What are the values of f_c and Δf at point a?
- Estimate the bandwidth of the FM signal. Assume the bandwidth of $m(t)$ to be the third harmonic frequency of $m(t)$.